



Final Presentation

Noisy Chord Sequence Denoising

Sai Dore and Kiran Dsouza

Problem Setup

Idea

Exploit temporal dependencies in chord progressions to recover corrupted labels more effectively

Inputs & Outputs

In: Noisy chord label sequence

Out: Denoised estimate of sequence

Chord Labels

Temporally dependent; denoising should use sequence structure

I	ii	iii	IV	V	vi	vii ^o
Major	Minor	Minor	Major	Major	Minor	Diminished
A	B	C#	D	E	F#	G#
B	C#	D#	E	F#	G#	A#
C	D	E	F	G	A	B
D	E	F#	G	A	B	C#
E	F#	G#	A	B	C#	D#

Central Question

When is explicit structured inference strongest, and when does learned message passing become more competitive?

Technical Framework

HMM Factorization

$$P(y_{1:T}, x_{1:T}) = P(y_1) \prod_{t=2}^T P(y_t | y_{t-1}) \prod_{t=1}^T P(x_t | y_t)$$

Hidden states (y)

Clean chord labels $y_1, \dots, y_T$

Observations (x)

Noisy chord labels $x_1, \dots, x_T$

Implementation Strategy

- **HMM:** Uses transition/emission models with Viterbi and forward-backward inference
- **GNN:** Retains chain structure, learning message-passing updates from data

Methods & Experimental Setup

Methods Compared

- No denoise baseline
- Spike removal
- Majority filter
- HMM (Viterbi & Marginal)

GNN-U: Trained on uniform noise

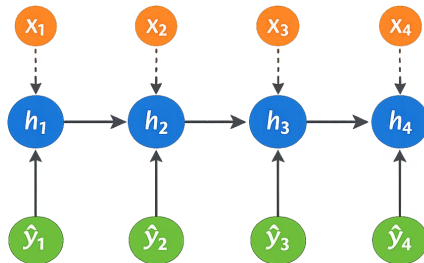
GNN-B: Trained on burst noise

GNN-Mix: Trained on both noise types

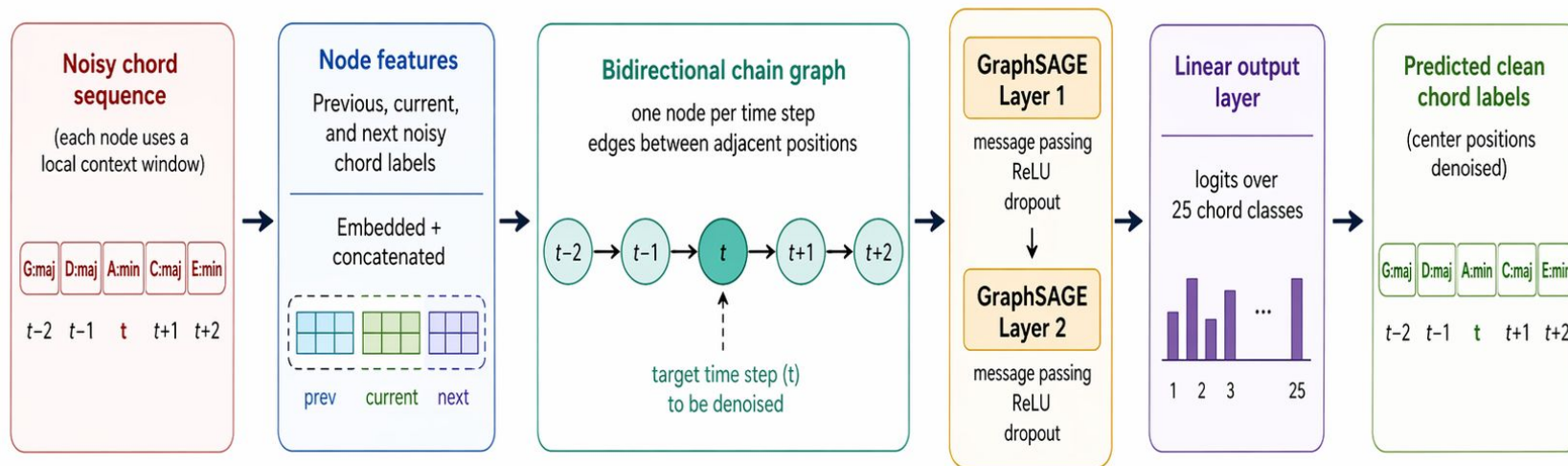
Setup Details

- **Dataset:** McGill Billboard (25 classes)
- **Data:** Beat-level chord sequences
- **Noise:** Uniform & Burst models
- **Metrics:** Accuracy & Change Rate

GNN for Chord Sequence Denoising



GNN Architecture



Note: Same architecture for GNN-U, GNN-B, and GNN-MIX; only the training noise differs.



Training objective: cross-entropy loss on noisy-clean sequence pairs.

Accuracy Under Uniform Noise

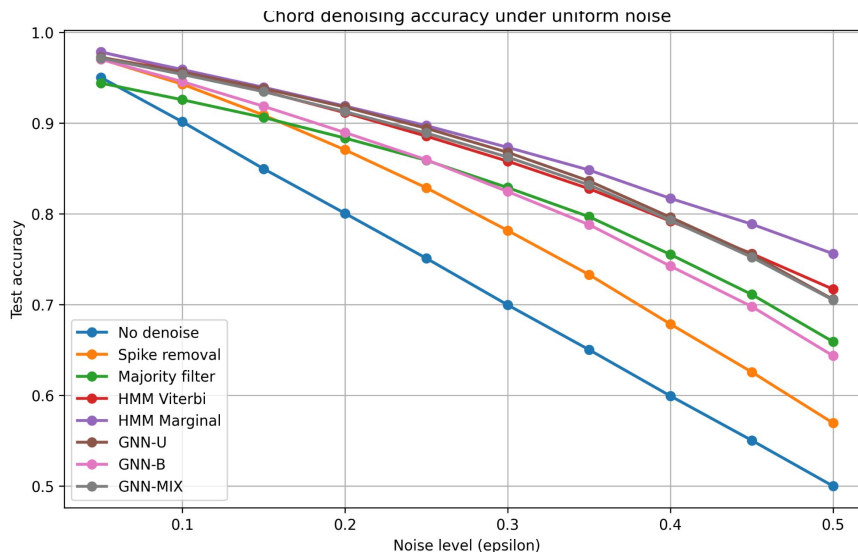
Key Findings

HMM methods perform best across most uniform-noise settings.

GNN-U and GNN-MIX are the strongest learned models in this matched setting.

At $\epsilon=0.50$:

- HMM-Marginal: **0.756**
- HMM-Viterbi: **0.717**
- GNN-U: **0.706**
- GNN-MIX: **0.705**
- No denoise: **0.500**



"Structured inference is strongest when the corruption process matches the HMM observation model."

Accuracy Under Burst Noise

Key Findings

Learned models become more competitive under burst noise.

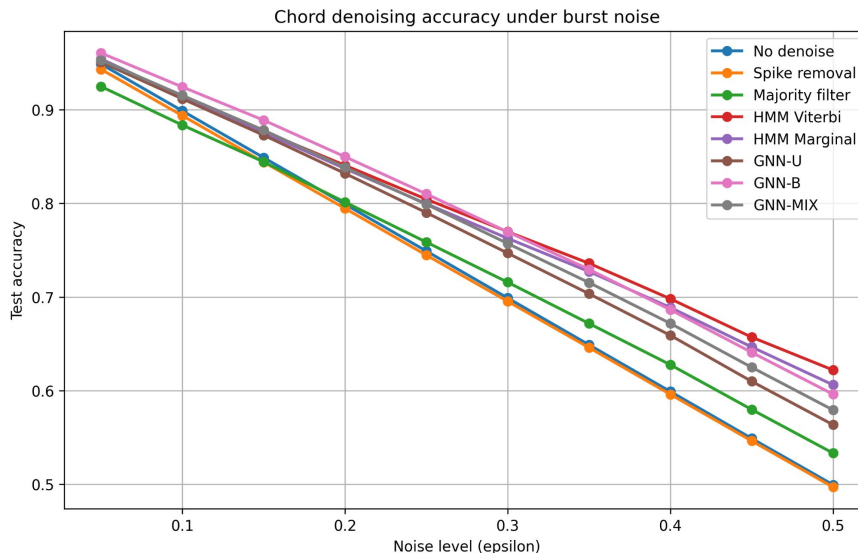
GNN-B is the strongest learned model, outperforming HMM at low/moderate noise.

Examples (ϵ Accuracy):

- $\epsilon=0.05$: GNN-B 0.961, MIX 0.953
- $\epsilon=0.10$: GNN-B 0.924, MIX 0.915
- $\epsilon=0.25$: GNN-B 0.810, MIX 0.799

At $\epsilon=0.50$:

- HMM-Viterbi: **0.622**
- HMM-Marginal: **0.606**
- GNN-B: **0.596**



"GNN-B is the strongest learned model and outperforms the HMM at several low and moderate noise levels."

Change Rate Results

Key Findings

Lower change rates indicate more stable chord predictions.

At $\epsilon=0.50$:

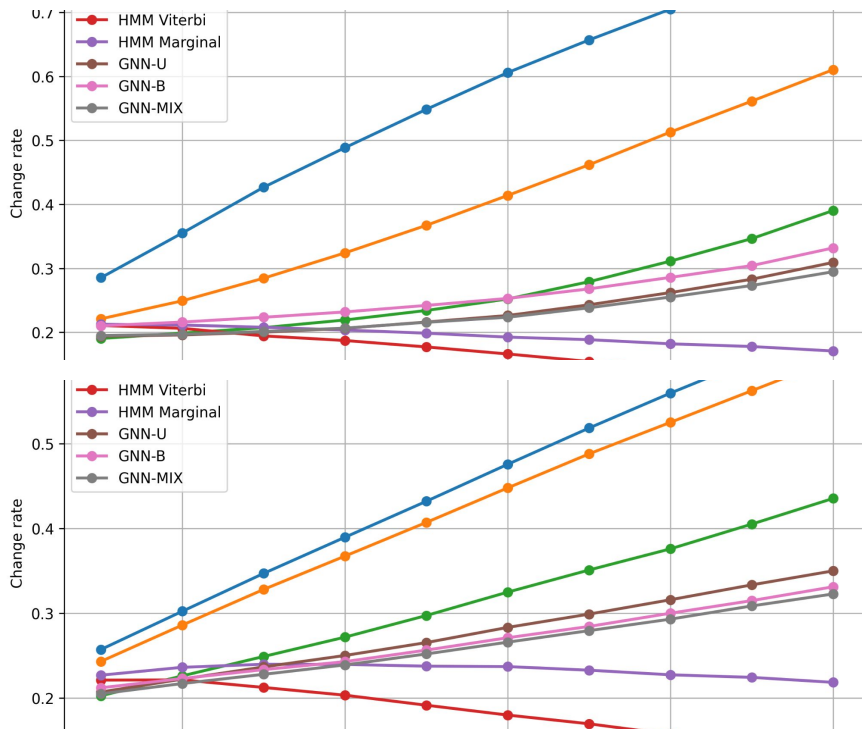
Uniform Noise

- No denoise: **0.788**
- HMM-Viterbi: **0.110**
- GNN-MIX: **0.295**

Burst Noise

- No denoise: **0.639**
- HMM-Viterbi: **0.127**
- GNN-B: **0.331**

Stronger denoisers are not only more accurate, but also more stable.



Examples

Setting	Example	Clean snippet	Noisy snippet	HMM-Marginal	Best GNN	Snippet accuracies
Uniform noise ($\epsilon=0.30$)	song 272 span 233:253	F#:maj across all positions	corrupted labels incl. G:maj,C#:min,	recovers clean snippet	GNN-MIX, with 1 error	Noisy = 0.40, HMM-M = 1.00, GNN-MIX = 0.95
Burst noise ($\epsilon=0.30$)	song 52, span 325:345	N repeated across all positions	corrupted burst incl. A:maj, D:min	recovers clean snippet	GNN-B, with 1 error	Noisy = 0.25, HMM-M = 1.00, GNN-B = 0.95

Under uniform noise, the HMM often gives the strongest correction

Under burst noise, GNN-B is often much closer to the HMM and can be nearly exact on corrupted spans

Limitations & Future Work



Limitations

- **Synthetic Noise:** Used instead of real-world human annotation noise.
- **Chord Vocabulary:** Limited to a simplified set of 25 classes.
- **Architecture:** Modest GNN design to isolate and focus on comparative analysis.



Future Work

Strategic directions for evolving the research:

- Explore richer, complex chord vocabularies.
- Develop more realistic, data-driven noise models.
- Investigate advanced architectures for robust sequence denoising.

Main Takeaways

Temporal Structure

Essential for effective chord-sequence denoising performance.

HMM Inference

Strongest in the matched uniform-noise setting.

Learned Message Passing

Much more competitive under burst noise scenarios.

Model Specialization

GNN-U (Uniform), GNN-B (Burst), GNN-MIX (Balanced).

Overall Conclusion

- Structure wins in matched settings
- Learning becomes more valuable under mismatch

AI Statement of Use



Coding and Organization Support

AI tools were utilized to provide essential support for both technical implementation and structural organization throughout the project lifecycle.

Key Contributions:

- Structuring the Graph Neural Network (GNN) implementation
- Organizing model code and clarifying message-passing pipelines
- Improving overall code readability and documentation